

Mid-infrared frequency combs

Albert Schliesser^{1,2}, Nathalie Picqué^{2,3,4*} and Theodor W. Hänsch^{2,3}

Laser frequency combs are coherent light sources that emit a broad spectrum of discrete, evenly spaced narrow lines whose absolute frequency can be measured to within the accuracy of an atomic clock. Their development in the near-infrared and visible domains has revolutionized frequency metrology while also providing numerous unexpected opportunities in other fields such as astronomy and attosecond science. Researchers are now exploring how to extend frequency comb techniques to the mid-infrared spectral region. Versatile mid-infrared frequency comb generators based on novel laser gain media, nonlinear frequency conversion or microresonators promise to significantly expand the applications of frequency combs. In particular, novel approaches to molecular spectroscopy in the 'fingerprint region', with dramatically improved precision, sensitivity, recording time and/or spectral bandwidth may lead to new discoveries in the various fields relevant to molecular science.

The mid-infrared spectral region of 2–20 μm (500–5,000 cm^{-1}) is a domain of interest to many areas of science and technology. A large number of molecules undergo strong characteristic vibrational transitions in this domain (Fig. 1a), thus making mid-infrared spectroscopy a univocal way to identify and quantify molecular species, including isotopologues, in a given environment. Mid-infrared spectroscopy provides not only a powerful tool for understanding the structure of molecular matter and its governing physical laws, but also for performing non-intrusive diagnostics of composite systems of physical, chemical or biological interest, in the gas, liquid or solid phase. The mid-infrared region also contains two important windows (3–5 μm and 8–13 μm) in which the Earth's atmosphere is relatively transparent. These regions can be exploited to detect small traces of environmental and toxic vapours down to sensitivities of parts-per-billion in a variety of atmospheric, security and industrial applications. The low Rayleigh scattering losses in the mid-infrared region benefit tomography and imaging in turbid media.

Since their introduction in the late 1990s, laser frequency combs have revolutionized precise measurements of frequency and time. The regular pulse train of a mode-locked femtosecond laser can give rise to a regular comb spectrum of millions of laser modes with a spacing equal to the pulse repetition frequency. Optical frequency combs in the visible and near-infrared domains have enabled the development of new ultraprecise optical atomic clocks, and commercially available combs have quickly matured to become standard instruments of precision spectroscopy. Frequency combs are now becoming enabling tools for an increasing number of applications, ranging from the calibration of astronomical spectrographs to molecular spectroscopy. The extension of frequency comb techniques to new spectral regions from terahertz frequencies to the extreme-ultraviolet is likely to trigger new applications of laser frequency combs. In the mid-infrared range, the advent of femtosecond optical combs is expected not only to bring a new set of tools for precision spectroscopy, but also to allow spectroscopy to be explored in a broader sense. A thorough understanding of molecular structure and dynamics often involves detailed spectral analysis over a broad frequency range. Precise investigations of changes in the composition of a molecular sample over a large dynamic range might also soon be within reach thanks to novel mid-infrared frequency-comb-based techniques.

This Review presents a summary of the advances in the emerging field of mid-infrared frequency comb technology and its

applications. We focus explicitly on mid-infrared sources, whose comb structure has been established, even though major achievements have also been reported in the development and exploitation of other broadband mid-infrared coherent sources. Although mid-infrared lasers and photonic technologies have been studied for over five decades, technical challenges have slowed down the development of convenient instrumentation in this spectral region. Over the past few years, in contrast, remarkable progress has been made in efficient mid-infrared frequency comb generators. A variety of innovative solutions — including novel laser gain media, nonlinear frequency conversion and Kerr comb generators — offer a wide choice of comb sources covering vast ranges of repetition frequency and spectral span. Compelling demonstrations in precision spectroscopy and direct frequency comb spectroscopy of molecules in the mid-infrared add enhanced capabilities to spectroscopy and provide new uses for frequency comb generators. Several new avenues of research are being opened by the introduction of the mid-infrared frequency comb.

Frequency comb generators in the mid-infrared

The characteristics of a frequency comb. A frequency comb^{1–6} is traditionally generated by means of a mode-locked laser: every time the pulse circulating in the laser cavity reaches the output coupler, a fraction of its energy is coupled out of the laser. The emitted pulse train therefore has a strictly periodic envelope function with a repetition frequency (f_{rep}) given by the inverse round-trip time of the pulse in the laser cavity $f_{\text{rep}} = v_g/L$, where v_g is the group velocity of the light in the cavity and L is the round-trip length of the cavity. However, due to dispersion in the laser resonator, the carrier wave of the pulse propagates at a phase velocity v_p , which is generally different from v_g . Therefore, the electric field of the outcoupled pulses is shifted with respect to the pulse envelope by a constant amount $\Delta\phi$ from pulse to pulse. This means that the electric field is not strictly periodic in time. Nevertheless, the Fourier spectrum of the electric field is composed of a discrete set of equidistantly spaced, sharp frequency components ν_n that obey the simple relation $\nu_n = n f_{\text{rep}} + f_{\text{ceo}}$, in which the carrier-envelope offset frequency is given by $f_{\text{ceo}} = \Delta\phi f_{\text{rep}}/2\pi$ and n is a large integer (Fig. 1b).

Mode-locked lasers. Mid-infrared radiation with such spectral properties can be generated through several techniques. Mode-locked lasers in the mid-infrared are similar to established sources

¹École Polytechnique Fédérale de Lausanne, CH-1015 Lausanne, Switzerland. ²Max-Planck Institut für Quantenoptik, Hans-Kopfermann Strasse 1, D-85748 Garching, Germany. ³Ludwig-Maximilians-Universität München, Fakultät für Physik, Schellingstrasse 4/III, 80799 München, Germany. ⁴Institut des Sciences Moléculaires d'Orsay, Centre National de la Recherche Scientifique, Bâtiment 350, Université Paris-Sud, 91405 Orsay, France.

*e-mail: nathalie.picque@mpq.mpg.de

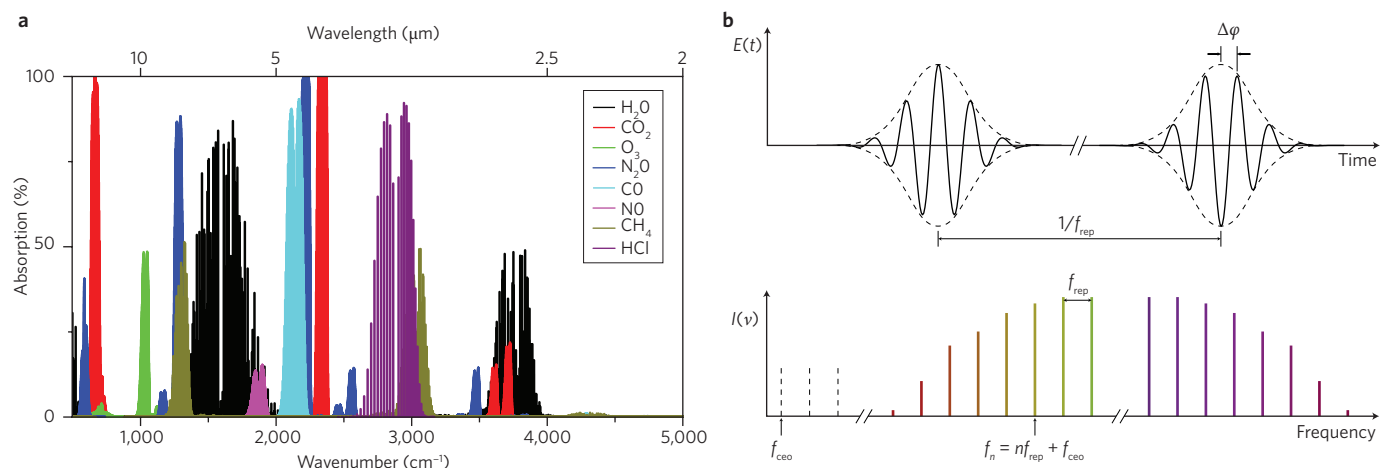


Figure 1 | Molecular mid-infrared fingerprints and the principle of the laser frequency comb. **a**, Absorption at the line centre for the main isotopologue of various gaseous molecules in the wavelength region of 2–20 μm. The calculation assumes a Doppler profile with a pressure of 133 Pa, an absorption path length of 1 cm and a temperature of 296.15 K. The individual line intensities are taken from the 2008 edition of the HITRAN database¹⁰⁷. **b**, Time-domain representation of a frequency comb pulse train together with the corresponding frequency-domain representation, which illustrates the discrete structure of the comb lines.

in the near-infrared (Fig. 2a). Transition metals such as chromium or iron provide a broad gain bandwidth in the mid-infrared region if doped into crystals of group II–VI compounds^{7–10}. Electronic transitions in these laser-active metals are strongly coupled to phonons in the lattice of the host crystal, which leads to significant homogeneous broadening — often more than 30% of the centre frequency. Chalcogenide hosts such as ZnSe and ZnS are particularly well-suited to this process because they support this ‘vibronic’ coupling, while sufficiently suppressing undesirable non-radiative multiphonon relaxation — even at room temperature — due to their low maximum phonon energy.

Using semiconductor saturable absorber mirrors or the Kerr lens effect, researchers have demonstrated mode-locked operation at repetition rates of around 100 MHz in Cr²⁺:ZnSe (refs 11–13) and Cr²⁺:ZnS (ref. 14) lasers, emitting relatively wide (>300 cm^{−1}) spectra around a wavelength of 2.5 μm and with powers of the order of 10 μW per mode (Figs 3, 4a). The comb structure of these lasers has been indirectly established by their successful application to dual-comb spectroscopy¹⁵ (see below). Further engineering could give access to the full ~2,000 cm^{−1}-wide gain bandwidth of chromium ions at 2.5 μm. Within the 4–5 μm region, tunable lasing has been demonstrated in Fe²⁺-doped crystals^{16–18}. Such lasers, if mode-locked, could provide frequency comb sources deep in the mid-infrared. Thulium, holmium and erbium ions also have several gain bands in the mid-infrared region¹⁹ (Fig. 3b). Thulium-doped silica fibre lasers emit at around 1.9 μm, and mode-locking has been achieved through a variety of techniques^{20–24}. Researchers have recently demonstrated a spectral coverage of 190 cm^{−1} with such a laser²⁵ (Fig. 3). In addition, powerful amplifiers based on doped fibres^{25,26} or crystals²⁷ can boost the power of these combs to the watt level, thus facilitating their additional spectral broadening in nonlinear fibres²⁵. Such fibre-based systems may quickly evolve into turn-key alignment-free instruments.

Difference-frequency generators. Laser sources have remained scarce at wavelengths beyond 3 μm. Quantum cascade lasers²⁸ can be tailored to operate at nearly arbitrary wavelengths, and recent experiments indicate that mode-locking is possible in principle^{29,30}. However, the comb properties of these rather narrow (<50 cm^{−1}) emission spectra have not yet been tested. In most cases, a different route is taken to access the spectral region beyond 3 μm, in which nonlinear optical effects are harnessed to transfer electromagnetic

energy from the visible or near-infrared domain into the mid-infrared (Fig. 2b,c).

For example, all modes of a near-infrared frequency comb can simultaneously be subjected to difference-frequency generation³¹ (DFG). When mixed with a continuous-wave (CW) laser of frequency ν_{CW} , the modes of a comb with frequencies $\nu_n = n f_{\text{rep}} + f_{\text{ceo}}$ can correspondingly be transferred to $\nu_n^{\text{DFG}} = |\nu_n - \nu_{\text{CW}}|$, which can lie in the mid-infrared for an appropriately chosen ν_{CW} . The efficiency of this process depends not only on the strength of the optical nonlinearity, but also on the ability to achieve phase-matching — that is, constructive interference of the radiation generated at different depths in the nonlinear crystal — over the wide necessary bandwidth. For broadband conversion, a common approach is to use periodically poled crystals, in which the sign of the nonlinear coefficient is reversed just at the depth where the generated waves would start to oscillate out of phase. This ensures that the difference-frequency builds up coherently all along the crystal. However, even in this case, imperfect phase-matching is often the limiting factor for the achieved bandwidth because the poling period is optimized for only part of the comb spectrum. This approach^{32,33} typically achieves a spectral width of 200 cm^{−1} and powers of the order of nanowatts per mode (Fig. 3). DFG between two synchronized combs has so far been marginally investigated³⁴, owing to its complexity and cost. Alternatively, difference-frequencies can be generated between the different teeth of a single comb. In this case $\nu_{n,m}^{\text{DFG}} = |n - m| f_{\text{rep}}$, where both n and m are integers. The comb’s carrier-envelope offset frequency is fixed to zero, which simplifies control over the modes’ absolute frequency. Using periodically poled lithium niobate (PPLN) as a nonlinear medium, researchers have generated spectra that span 3–5 μm with a width of more than 500 cm^{−1} and powers of up to 10 nW per comb mode^{35–38}. At a wavelength of around 5 μm, absorption in lithium niobate sets a limit on how far this approach can be pushed into the mid-infrared. Other materials³¹ must be used to access longer wavelengths (Fig. 3b). For example, 300-cm^{−1}-wide spectra tunable in the range of 3–17 μm have been generated in gallium selenide, with powers of up to 50 nW per comb tooth^{39–43} (Fig. 4b).

Optical parametric oscillators. A key challenge in sources based on nonlinear optics is the efficiency of the desired photon-conversion process. This metric often remains well below 1%, even in the best

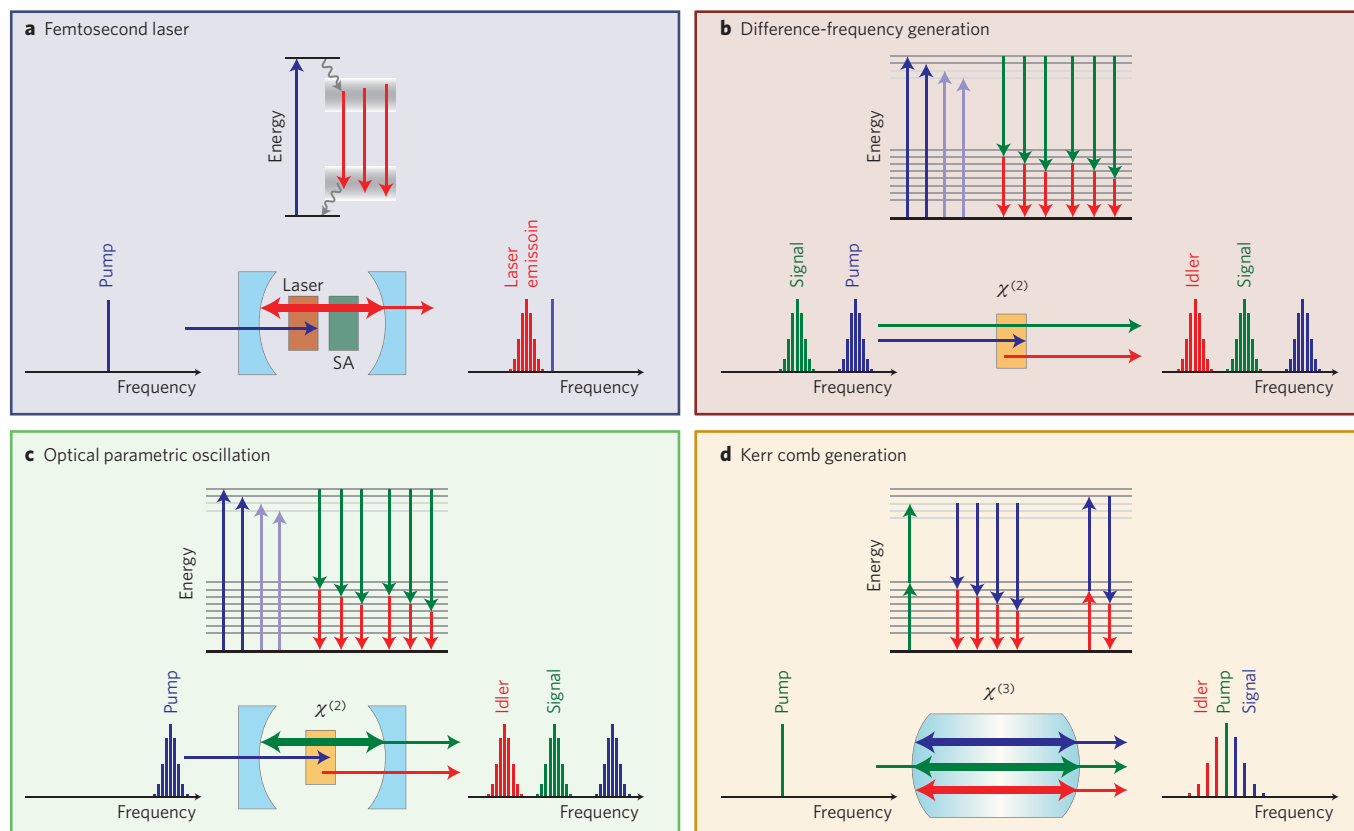


Figure 2 | Approaches to frequency comb generation in the mid-infrared. **a**, Femtosecond mid-infrared laser. A laser-active medium with broad gain is inverted by a strong optical pump field. In a resonator that additionally holds an (effective) saturable absorber (SA), a short laser pulse builds up and circulates in the cavity. The spectrum of the outcoupled laser light constitutes a frequency comb. **b**, DFG. In a medium with a second-order nonlinearity $\chi^{(2)}$, nonlinear frequency mixing of the multitude of modes of two ‘signal’ and ‘pump’ combs leads to the generation of various difference-frequencies, forming an ‘idler’ comb in the mid-infrared. Alternatively, either the signal or pump can be a single CW laser. **c**, Optical parametric oscillation. An $\chi^{(2)}$ -nonlinear medium placed in an optical resonator provides gain to the cavity’s ‘signal’ modes if pumped by a strong pump beam. Coherent oscillation — seeded only by the intracavity quantum fluctuations — sets in if this gain exceeds the losses of the signal modes. The desired mid-infrared idler modes are concomitantly generated due to energy conservation. **d**, Kerr comb generation. A dielectric resonator made of a $\chi^{(3)}$ -nonlinear material is pumped by a mid-infrared CW source. Signal and idler sidebands are generated in the modes of the resonator adjacent to the pump mode through degenerate four-wave mixing. Due to the high quality factor of the modes, this process can be very effective and thus cascade to a large number of sideband modes. In addition, non-degenerate four-wave mixing can lock the phases of all generated sidebands.

circumstances³⁶. One way of boosting efficiency is to use an optical parametric oscillator (OPO, Fig. 2c)^{44–46}. If pumped by a femtosecond laser, many longitudinal ‘signal’ modes of the resonator in an OPO can simultaneously experience gain that exceeds the threshold for parametric oscillation. Of course, achieving oscillation for the many signal and idler modes requires, in addition to phase matching, management of the resonator’s dispersion to ensure that the equidistant signal modes coincide with the resonator’s modes across the desired wide bandwidth. In the time domain, this corresponds to the requirement that the power of the signal mode remains concentrated in a short pulse that passes through the nonlinear medium synchronously with the pump pulses. Using PPLN-based OPOs pumped by compact solid-state⁴⁷ or powerful fibre-laser-based near-infrared combs⁴⁸, researchers have demonstrated wide ($>100\text{ cm}^{-1}$) spectra that are tunable over a range of 2–4 μm and have powers of up to 30 μW per mode.

Although the signal and idler combs inherit the mode spacing f_{rep} from the pumping comb, the carrier-envelope offset frequency of these combs is subject only to the condition $f_{\text{ceo}}^{\text{signal}} + f_{\text{ceo}}^{\text{idler}} - f_{\text{ceo}}^{\text{pump}} = n f_{\text{rep}}$, where $f_{\text{ceo}}^{\text{signal}}$, $f_{\text{ceo}}^{\text{idler}}$ and $f_{\text{ceo}}^{\text{pump}}$ are the carrier-envelope offset frequencies of the signal, idler and pump combs, respectively. For a fully stabilized pump source (f_{rep} and $f_{\text{ceo}}^{\text{pump}}$ fixed),

one degree of freedom must still be controlled. In many recent implementations of mid-infrared OPO sources, this is facilitated by the generation of optical sum and second-harmonic frequencies occurring concomitantly with the OPO action in the nonlinear crystal^{46–48}, which provides control of $f_{\text{ceo}}^{\text{signal}}$ and therefore also over $f_{\text{ceo}}^{\text{idler}}$. In the special case of a degenerate OPO⁴⁹, in which the signal and idler combs merge into one comb of subharmonic frequencies, the additional constraint $f_{\text{ceo}}^{\text{signal}} = f_{\text{ceo}}^{\text{idler}}$ allows only the values $f_{\text{ceo}}^{\text{pump}}/2$ and $(f_{\text{ceo}}^{\text{pump}} + f_{\text{rep}})/2$ for the subharmonic comb’s carrier-envelope offset frequency, which further simplifies the task of stabilization. Such a degenerate OPO has recently enabled the generation of ultra-broad mid-infrared spectra spanning 900 cm^{-1} around a wavelength of 3 μm with an impressive power of 200 nW per mode⁵⁰. Another noteworthy route is the use of orientation-patterned GaAs as the nonlinear medium, as its transparency in the deep mid-infrared provides access to longer wavelengths^{51,52}.

Microresonator-based Kerr combs. Recently, researchers have reported that a special form of parametric oscillation (sometimes referred to as hyper-parametric oscillation⁵³) can be used to generate a frequency comb in an optical microresonator⁵⁴ (Fig. 2d). The resonator comprises a toroidal ring of sub-millimetre diameter that

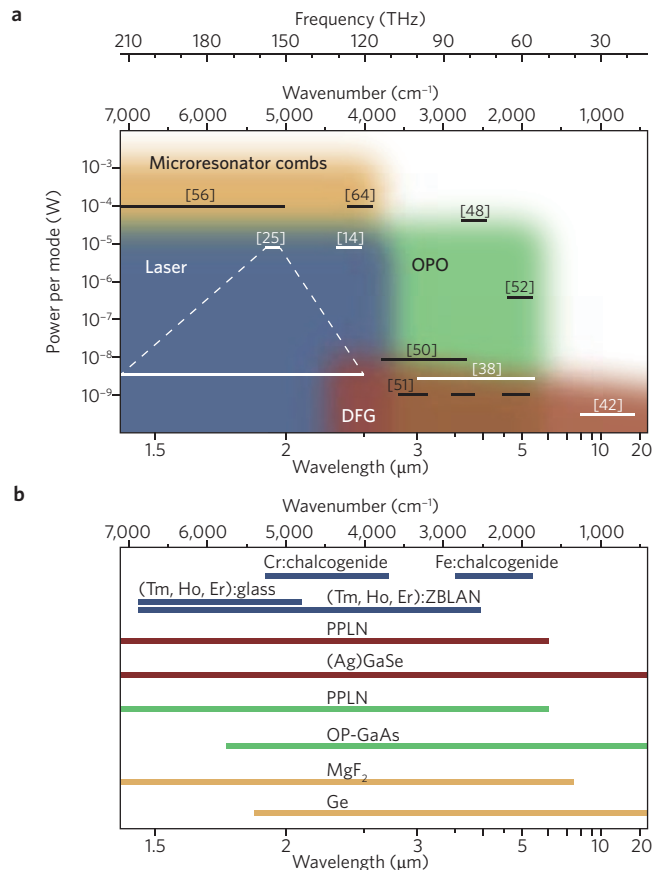


Figure 3 | Spectral coverage of different approaches and materials.

a, Overview of the spectral regions and power range accessible with mid-infrared femtosecond lasers (blue), DFG (red), OPOs (green) and microresonator-based Kerr combs (yellow). Black and white bars represent a selection of realized comb sources, with citations shown in brackets. The width of each bar represents the spectral span over which the respective comb has at least the power per mode indicated by its position on the ordinate. Note that many of the sources shown here are also wavelength-tunable (not shown). **b**, Potential spectral coverage of selected materials if employed in mid-infrared comb generators (generation approaches are colour-coded as in **a** and Fig. 2). Note that the range indicated for the rare-earth ions covers several different transition bands in each of the ions.

can host very-high-quality optical whispering-gallery modes. If populated with a strong pump field, the third-order nonlinearity $\chi^{(3)}$ of the dielectric resonator material can lead to four-wave mixing. For a single-frequency CW pump, this implies the generation of a pair of signal and idler photons, whose combined energy corresponds to the energy of two pump photons, $h\nu_{\text{signal}} + h\nu_{\text{idler}} = 2h\nu_{\text{pump}}$. Owing to the high finesse of the resonators (up to 10^7), the threshold for this process can be particularly low if all involved fields are resonant with the cavity modes. This is possible if dispersion and nonlinear phase shifts due to self- and cross-phase modulation are appropriately balanced⁵⁵. Remarkably, a single-frequency pump laser can give rise not only to one signal-idler pair, but also to a massive cascade of signal and idler sidebands. These sidebands become mutually phase-locked by non-degenerate four-wave mixing, thus only requiring $h\nu_{\text{signal}} + h\nu_{\text{idler}} = h\nu'_{\text{signal}} + h\nu'_{\text{idler}}$. The resulting 'Kerr' comb can cover an entire octave^{54,56,57} and extend into the mid-infrared if pumped by a strong near-infrared laser. A similar mechanism is at work in the generation of mid-infrared 'super-continua', which is reported to occur in waveguides made from

infrared-enabled materials. However, recent experiments^{58–61} have so far not shown that the coherence of the resulting spectra is not destroyed by competing nonlinearities.

Resonators made from crystalline materials such as calcium and magnesium fluoride support ultrahigh-Q whispering-gallery modes deep in the mid-infrared ($\lambda < 7 \mu\text{m}$) and simultaneously exhibit the anomalous dispersion required for efficient and low-noise comb generation^{62,63}. Researchers have employed pumping at $2.45 \mu\text{m}$ to demonstrate a Kerr comb covering the range of $2.35\text{--}2.5 \mu\text{m}$ (at around -60 dB)⁶⁴ (Fig. 4d). The combination of a powerful pump source and a large mode spacing of 107 GHz (corresponding to the round-trip along the resonator's perimeter) enables milliwatt powers per mode (Figs 3, 4d). This novel approach enables the generation of simple and compact mid-infrared combs with yet unexplored opportunities for different resonator materials (Si, Ge and InP) or pump sources. Finally, the large mode spacing could be beneficial for a number of emerging applications in spectroscopy and metrology, as described below.

Applications

The first applications of mid-infrared frequency combs concentrated on the precision spectroscopy of molecules. Recent experiments have demonstrated an intriguing potential for the rapid and sensitive acquisition of molecular spectra over a vast spectral span. Foreseen applications might push the frontiers of spectroscopy and dynamics even further by employing coherent control or nonlinear phenomena.

Frequency combs as spectroscopic calibration tools. Although frequency combs were initially conceived for precision spectroscopy of the hydrogen atom, they are now commonly used for precise frequency measurements of atomic resonances and molecular rovibronic transitions throughout the visible and near-infrared spectral regions. The mid-infrared spectral region offers additional opportunities due to the presence of strong rovibrational transitions in a number of molecules (Fig. 1a). Accurate frequency measurements in the mid-infrared provide an in-depth understanding of a molecule's energy levels. Because frequency combs can conveniently link optical and microwave frequencies, they provide the long-missing mechanism needed to realize an optical clock. In the mid-infrared region, frequency combs have enabled the demonstration of molecular optical clocks, using, for example, the hyperfine $F_2^{(2)}$ optical transition^{35,37} of the P(7) line of the ν_3 band in methane as a pendulum with a fractional instability of 1.2×10^{-13} . This allows the comparison between a mid-infrared clock and other optical clocks.

Comb-enabled accurate frequency measurements (Fig. 5a) in the mid-infrared also allow new tests of fundamental physical laws and improvements in the determination of fundamental constants, as well as sensitive limits for their possible slow variation. For instance, the frequency comparison of a rovibrational transition in SF_6 (Fig. 5b) around $10 \mu\text{m}$ — accessed through spatial two-photon Ramsey fringes — to a caesium microwave atomic clock by means of a frequency comb and an optical link⁶⁵ has set an upper limit on a possible small variation of the proton-to-electron mass ratio⁶⁶. These model-free results are derived from an absolute frequency measurement within an impressive fractional uncertainty of 2.8×10^{-14} . Such data are complementary to those retrieved from astronomical observations, where measurements are effectively separated in time by several billion years.

Precise measurements of the Boltzmann constant have so far been performed using acoustic gas thermometry, with an inaccuracy of 1.7×10^{-6} . In recent years, researchers have accurately measured⁶⁷ the Doppler profile of a well-isolated absorption line of a molecular gas as a potential way to determine the Boltzmann constant by optical means. This requires, in addition to in-depth

line-shape modelling and control over systematic effects, accurate control of the frequency of the laser that probes the absorption profile. As a last example, measuring small differences between the infrared transition frequencies of the two enantiomers of a chiral molecule might provide evidence⁶⁸ on parity violation due to the weak interaction. However, the latest predictions estimate such relative differences to be as small as 8×10^{-17} for the CHFCIBr molecule, and of the order of 10^{-13} – 10^{-14} for oxorhenium complexes around $10 \mu\text{m}$ in size. The latter sensitivity, although technically challenging for such complex molecules, might be soon within reach using a state-of-the-art ultra-high-resolution set-up⁶⁶.

Precision spectroscopy in the mid-infrared uses the frequency comb as a frequency ruler (Fig. 5a), similar to the way it is used in the near-infrared or visible regions. The unknown absolute frequency ν_i of a CW laser is determined by creating a beat note f_b , thus giving $\nu_i = n f_{\text{rep}} + f_{\text{ceo}} \pm f_b$. The correct sign of f_b is determined by a small change in either of the frequencies, and the mode number n is determined by a coarse measurement of ν_i , for example, with a wavemeter. Although researchers have successfully referenced a CW laser to a mid-infrared frequency ruler, such experiments have not yet reached the performance of their visible counterparts. Frequency combs have been produced by DFG for the direct referencing of methane-stabilized HeNe lasers^{35,37} at $3.39 \mu\text{m}$ or DFG CW sources⁶⁹ at $3 \mu\text{m}$. The main drawback of this approach is that both the comb and the probing laser emit in the mid-infrared region, where low-noise high-speed detectors — and more generally advanced photonics technology — are lacking. Upconversion of the infrared light from the CW laser is therefore a preferred alternative. Sum-frequency generation between the CW laser and the near-infrared comb produces a frequency-shifted comb that may still overlap the initial comb. The resulting phase-coherent beating signal can be used to determine the frequency of the CW laser independently of the comb's carrier-envelope offset frequency. Such an approach has been implemented in, for example, Doppler-free two-photon spectroscopy of SF_6 and saturated-absorption spectroscopy of formic acid with CO_2 lasers around 9 – $10 \mu\text{m}$ (refs 70,71) and Doppler-limited spectroscopy of CO_2 with quantum cascade lasers at $4.3 \mu\text{m}$ (ref. 72). The metrological link to a near-infrared frequency comb could also be realized⁷³ by sum-frequency generation between two CW lasers, as demonstrated, for example, by mixing a $4.4 \mu\text{m}$ quantum cascade laser and a $1.064 \mu\text{m}$ Nd:YAG laser. Alternatively, when the infrared radiation of the CW probing laser is produced by the nonlinear frequency downconversion of near-infrared lasers, the near-infrared CW laser may be referenced to a near-infrared frequency comb, as already implemented in schemes involving DFG^{74,75} (Fig. 5c) and optical parametric conversion⁷⁶.

Direct frequency comb spectroscopy. Direct frequency comb spectroscopy, a denomination that first appeared in 2004⁷⁷, was the first application of frequency combs. In the mid-1970s, researchers showed that picosecond mode-locked dye lasers^{78,79} can produce a stable phase-coherent pulse train, and then used it to achieve direct excitation of atomic vapours. Direct frequency comb spectroscopy of atomic resonances by Ramsey-like excitation with a coherent train of multiple light pulses and Doppler-free two-photon excitation therefore has a long history in the visible and ultraviolet regions. This technique involves monitoring the fluorescence from an excited sample while tuning either the repetition frequency or the carrier-envelope offset. The comb can then be used in a manner similar to a CW laser, with the added advantage that the high peak intensity of pulsed lasers facilitates efficient nonlinear conversion into frequency regions where CW lasers are not yet available.

In recent years, novel techniques^{80–85} have been developed in which the comb directly interrogates a vast number of transitions

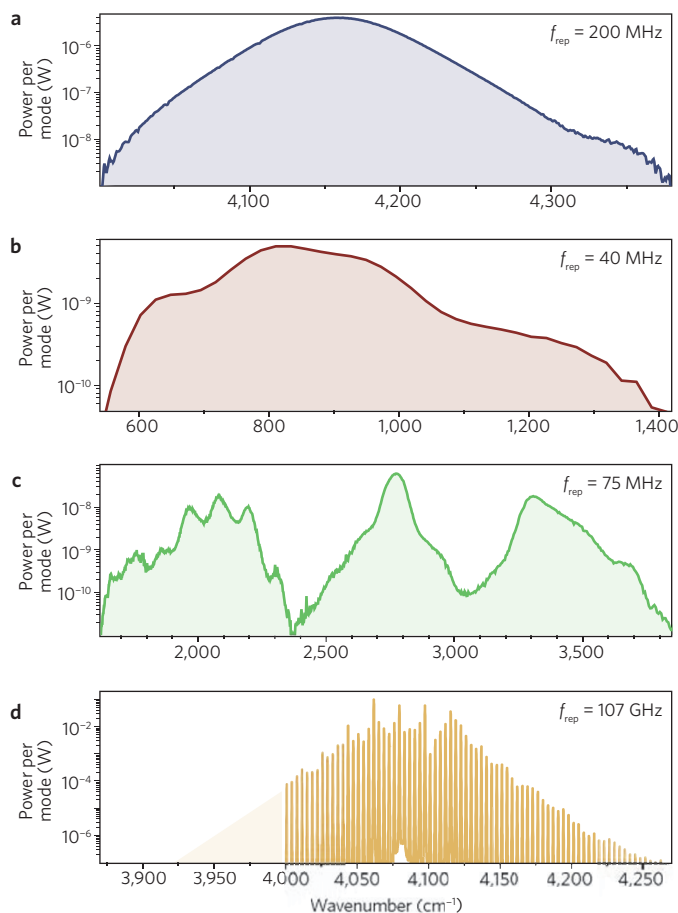


Figure 4 | Typical mid-infrared spectra obtained through different approaches. a, Femtosecond Cr:ZnSe laser¹¹. **b**, DFG in GaSe (ref. 42).

c, Optical parametric oscillation in GaAs (ref. 51). **d**, Kerr comb generation in a MgF_2 microresonator (ref. 64). Note that only in **d** could the individual comb components be resolved by the employed spectrograph. The shaded area in this panel indicates expected spectral components outside the spectrograph's measurement range.

of an absorbing sample. Direct frequency comb absorption spectroscopy of molecules in the gas phase has so far been the most prominent application of mid-infrared frequency combs. The mid-infrared domain, with its strong molecular fingerprints, attracts considerable interest. Direct absorption frequency comb spectroscopy can provide short measurement times, high sensitivity and high accuracy over a broad spectral bandwidth. The spectrum of the comb is frequency-selectively attenuated and phase-shifted by the molecular resonances, and is analysed using a spectrometer. So far, the preferred spectrometric approach has been Fourier transform spectroscopy⁸⁶ because the dispersers associated with detector arrays are neither efficient nor conveniently available in the mid-infrared. Fourier spectrometers record the spectroscopic signal on a single photodetector and therefore overcome dispersive-related issues. Two different implementations of Fourier transform spectrometers have been reported: Michelson-based Fourier transform spectroscopy, and dual-comb spectroscopy. Each presents its own distinct advantages, but both rely on the same physical principle (Fig. 6a,b). Because the frequencies of the mid-infrared comb are too high (15 – 150 THz) to be counted directly, these techniques downconvert the signal to a lower frequency region where it can be manipulated through digital signal processing. Frequency comb Fourier transform spectroscopy is

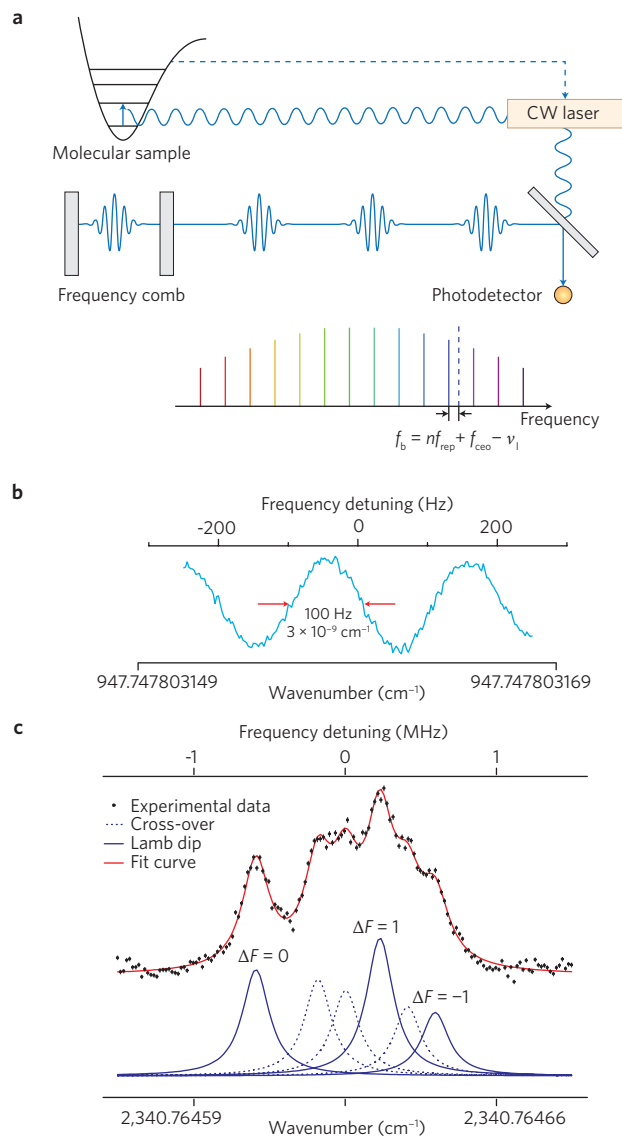


Figure 5 | Frequency comb as a frequency ruler. **a**, Laser frequency combs are used to phase-coherently compare optical and microwave frequencies in a single step. The frequency of any comb line can be calculated from the two radiofrequencies f_{rep} and f_{ceo} , which can be compared or phase-locked to a microwave reference, together with the integer mode number n . Optical frequency measurements are often used to determine the absolute frequency of an atomic or molecular transition interrogated by a single-frequency CW laser. By heterodyning the CW laser against a nearby optical comb line, the frequency of the CW laser, ν_i , can be measured. The resulting beat note f_b can then be counted with standard radiofrequency equipment. **b**, Comb-assisted ultrahigh-resolution mid-infrared spectroscopy⁶⁶. A spatial Ramsey interferometer resolves a rovibrational transition in the $2\nu_3$ band of SF_6 to within 100 Hz. The $10\text{ }\mu\text{m}$ CO_2 laser used to interrogate the two-photon resonance in a molecular beam is referenced⁷⁰ using a frequency comb and an optical link to a caesium fountain clock. The frequency measurement can be achieved within an uncertainty of 0.79 Hz ($2.6 \times 10^{-11}\text{ cm}^{-1}$). **c**, Comb-assisted Doppler-free ultrasensitive mid-infrared spectroscopy⁷⁴. Saturated-absorption cavity ring-down spectroscopy with a tunable CW laser is a technique that combines sensitivity, high resolution and frequency accuracy through frequency comb referencing. The hyperfine triplet and related crossovers of the $\text{R}(0)$ transition of the ν_3 band of the rare isotopologue $^{17}\text{O}^{12}\text{C}^{16}\text{O}$ around $4.27\text{ }\mu\text{m}$ is observed at a partial pressure of 150 μPa . With a rubidium/GPS-disciplined frequency comb, its line positions can be measured within $1.0 \times 10^{-7}\text{ cm}^{-1}$.

a time-domain technique in which the pulse train of a comb is interferometrically sampled — akin to an optical sampling oscilloscope — by a second pulse train of different repetition frequency, which may be produced either by the varying optical delay of a Michelson interferometer or by a second frequency comb source.

In Michelson-based frequency comb Fourier transform spectroscopy (Fig. 6c), the pulse train of a frequency comb with a pulse repetition frequency f_{rep} is analysed by a Michelson interferometer that records interference versus path difference $2\nu t$, where t is time and ν is the constant velocity of the moving mirror. The comb travelling along the fixed mirror arm keeps its native line-spacing f_{rep} . The comb in the moving arm of the interferometer has a Doppler-shifted repetition frequency of $f_{\text{rep}}(1 - 2\nu/c)$. The two interfering frequency combs exiting the interferometer interrogate the absorbing sample and beat on a photodetector. Due to the limited speed of the moving mirror, the optical frequencies are down-converted to the acoustic range of $<100\text{ kHz}$. Using a mid-infrared frequency comb — or any broadband coherent source — as the light source for a Michelson-based Fourier transform spectrometer dramatically reduces the measurement time (or improves the signal-to-noise ratio) owing to the high spectral radiance of the coherent source. This concept has been demonstrated using a Cr:ZnSe mode-locked laser¹¹ with a wavelength of around $2.4\text{ }\mu\text{m}$ and an OPO-based frequency comb⁸⁷ that is tunable in the range of $2.8\text{--}4.8\text{ }\mu\text{m}$ and has a width of $50\text{--}400\text{ cm}^{-1}$. Sensitivity to weak absorption may be greatly enhanced when the effective absorption pathlength is increased using a multipass cell or a high-finesse resonator, in which the optical cavity is matched to the laser resonator so that it is simultaneously resonant for many comb lines. This scheme therefore requires dispersion-managed cavity mirrors, control over the two degrees of freedom of the comb for coherent addition of the pulses circulating inside the resonator, and tight locking of the comb to the cavity to avoid frequency-to-amplitude noise conversion. The strong intensity of the rovibrational mid-infrared lines associated with a sensitivity enhancement in the cavity of several thousand provides⁸⁸ detection levels as low as one part per billion for H_2O_2 at a wavelength of around $3.75\text{ }\mu\text{m}$ (Fig. 6d). Another intriguing application explores the potential of broadband near-field microscopy with a frequency comb produced by DFG around $1,000\text{ cm}^{-1}$ to map interferograms of the near-field interaction between a metal tip and polar materials within a few seconds per tip-illuminated spot⁸⁹.

Michelson-based Fourier transform spectroscopy has been mastered for more than 40 years, so replacing the traditional incoherent light source with a frequency comb does not present major difficulties. Dual-comb spectroscopy, in contrast, is in its infancy and thus its full potential is yet to be fully realized.

In dual-comb spectroscopy (Fig. 6e), two frequency combs with slightly different line spacings are heterodyned after passing through the cell, yielding a downconverted radiofrequency comb that contains information on the absorption experienced by both combs. Other implementations allow a single comb to interrogate the sample and also to measure the dispersion induced by the sample. The static scheme of dual-comb spectroscopy overcomes the shortcomings induced by the moving mirror of the Michelson interferometer. Its resolution is limited only by the measurement time and the line spacing of the combs. Interleaving several spectra may improve the resolution down to the comb intrinsic linewidth. Moreover, the entire radiofrequency domain up to $f_{\text{rep}}/2$ can be covered, and is restricted only by aliasing. Recording times in dual-comb spectroscopy may be much less than those of Michelson-based Fourier transform spectroscopy. Much literature is available on the practical implementation of a dual-comb spectrometer^{82,92–94}.

Low-resolution, proof-of-principle demonstrations of dual-comb spectroscopy with difference-frequency generators have

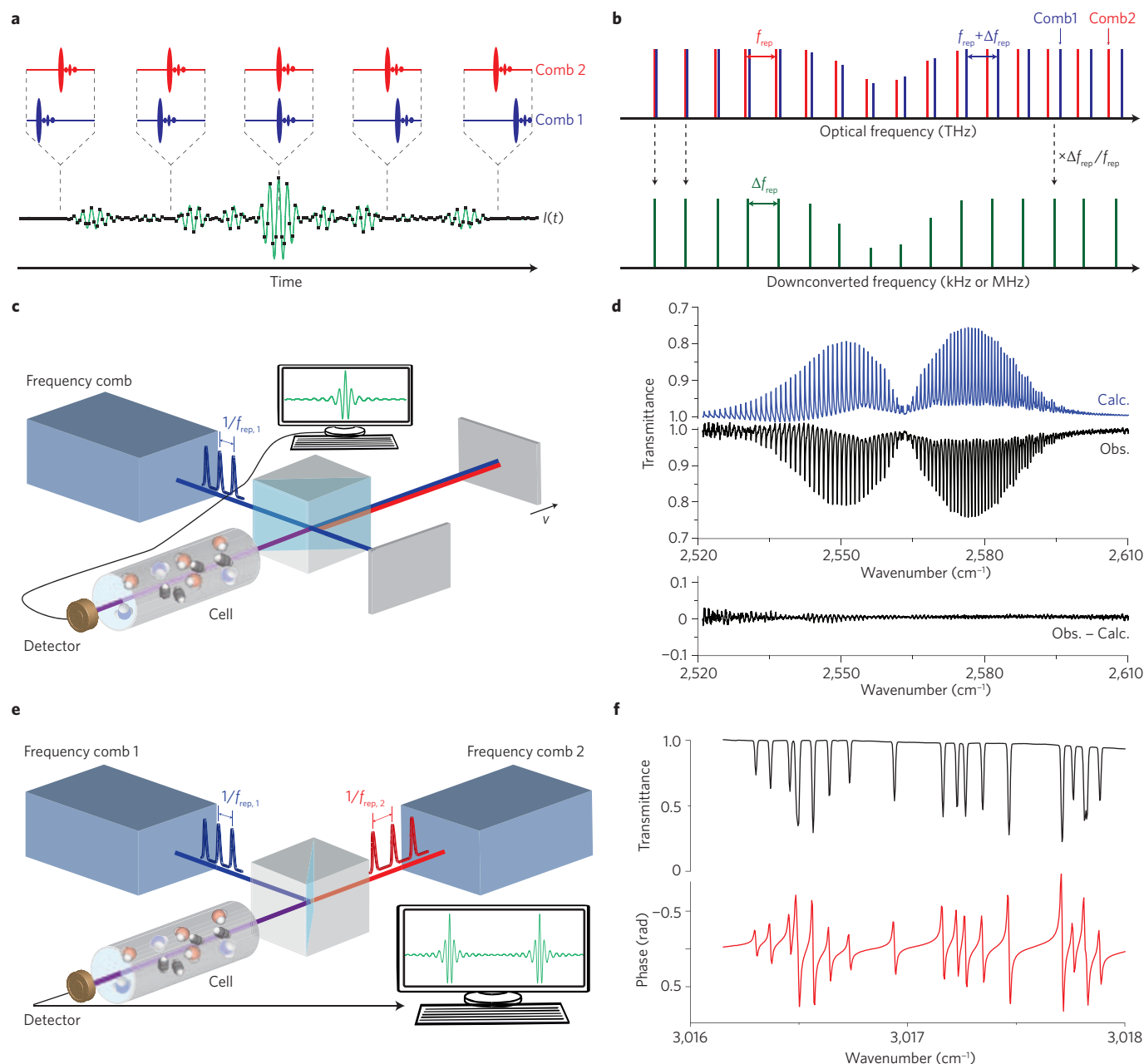


Figure 6 | Frequency comb Fourier transform spectroscopy. **a**, Time-domain picture of Fourier transform spectroscopy. The pulse train of comb 1 mixes with the pulses from comb 2 to generate a cross-correlation interferogram $I(t)$. **b**, Frequency-domain picture of Fourier transform spectroscopy. Owing to interference between the pairs of optical comb lines, the optical absorption spectrum is effectively mapped into the acoustic or radiofrequency region, where it becomes accessible to fast digital signal processing. The frequencies of the optical spectrum are downconverted to $\Delta f_{\text{ceo}} + n\Delta f_{\text{rep}}$, where Δf_{ceo} is the difference between carrier-envelope slippage frequencies of the two combs, n is an integer and Δf_{rep} is the difference between the comb repetition frequencies. **c**, Michelson-based frequency comb Fourier transform spectroscopy. **d**, Rotationally resolved experimental (obs.) and computed (calc.) spectra⁸⁸ of the $2\nu_1$ overtone band of N_2O , around $3.9\ \mu\text{m}$, by cavity-enhanced frequency comb Fourier transform spectroscopy at a resolution of $2.8 \times 10^{-2}\ \text{cm}^{-1}$. The enhancement resonator with a finesse exceeding 3,800 contains 0.8 ppm of nitrous oxide at 1,013 hPa of nitrogen. **e**, Dual-comb spectroscopy. **f**, Portion³³ of absorption and dispersion dual-comb spectra showing the Q(8)-Q(4) multiplets of the ν_3 band of methane. In ref. 33, each spectrum of methane (pressure of 27 Pa and absorption length of 28 cm) spans $33\ \text{cm}^{-1}$ and is measured, with a signal-to-noise ratio of 3,500, within 13 min at a resolution of $3 \times 10^{-3}\ \text{cm}^{-1}$. In total, the positions of 132 rovibrational transitions, spanning over $150\ \text{cm}^{-1}$, are determined within an uncertainty of 300 kHz.

been carried out at wavelengths of around $10\ \mu\text{m}$ since 2004, probing gaseous^{39,40} and solid-state⁹⁰ samples. Although limited in terms of optical power and comb stability, these demonstrations highlighted one promising characteristic of dual-comb spectroscopy: the very short measurement times enabled by the absence

of moving parts. Frequency-comb-based scanning near-field optical microscopy has also been explored⁹¹. The lack of efficient mid-infrared frequency comb sources and the difficulty associated with synchronizing the pulse trains of two combs with interferometric precision have hindered the development of mid-infrared

dual-comb spectroscopy, and most reports^{82,92–94} have therefore been in the near-infrared domain. Stimulated Raman dual-comb spectroscopy⁹⁵ is an alternative technique for accessing fundamental transitions with near-infrared combs. However, a recent proof-of-principle demonstration¹⁵ using unstabilized Cr:ZnSe mode-locked lasers around 2.4 μm enabled the rovibrational spectrum of acetylene in the range of 3,960–4,220 cm^{-1} to be recorded within 10 μs at a resolution of 12 GHz. DFG dual-comb experiments at around 3.4 μm have exhibited³³ (Fig. 6f) a high accuracy of 10^{-5} cm^{-1} when measuring the line positions of the Doppler-broadened transitions of the ν_3 band of methane. Simplification of the laser systems for dual-comb spectroscopy is also highly desirable; an intriguing option is offered by a 3.3 μm femtosecond OPO emitting two asynchronous trains of pulses⁹⁶, with no significant nonlinear coupling between the two channels.

Michelson-based frequency comb Fourier transform spectroscopy and dual-comb spectroscopy share a number of advantages over traditional spectrometric instruments: as with any Fourier transform spectrometer, the spectral span is limited only by the spectral bandwidths of the source and detector. The multiplex advantage grants overall consistency of the simultaneously measured spectral elements, which may prove crucial when, for example, investigating transient phenomena. The spatial directionality of laser sources permits remote measurements and microscopic focusing. The high spectral radiance of frequency comb sources provides high signal-to-noise ratios and reduced measurement times. The comb structure of the spectrum dramatically improves the detection sensitivity through the optimal use of cavity-enhanced techniques. Moreover, resolving the comb lines allows for self-calibration of the frequency scale of the spectra. Analysing the comb light with a Michelson interferometer results in a robust, fast and sensitive multiplex spectrometer, and the already high quality of the achievements guarantees immediate opportunities for many applications, such as in the trace gas detection of multiple species. Dual-comb spectroscopy is more technically challenging than Michelson-based spectroscopy and still requires considerable work to explore its full potential. However, the ability to record Doppler-limited 300 cm^{-1} mid-infrared spectra within 100 μs is clearly within reach — as are resolutions limited only by the comb linewidth. The absence of moving parts in dual-comb spectroscopy overcomes the speed and resolution limitations of the Michelson-based approach, and may lead to compact high-resolution instruments. One may even envision a chip-size dual-comb spectrometer based on microresonators for real-time spectroscopy in the liquid phase.

Combining frequency combs with other advanced tools in the fields of laser science, nonlinear optics and electronic signal processing may vastly enhance the range and capabilities of molecular laser spectroscopy. However, molecular physics with laser frequency combs is still in its early stages of development. Both Michelson-based frequency comb Fourier transform spectroscopy and dual-comb spectroscopy can be combined with, for example, Doppler-free measurement schemes, hyperspectral imaging, microscopy, temporal resolution or selective molecular detection techniques, while maintaining their already demonstrated capabilities. More interestingly, because laser frequency combs involve intense ultrashort laser pulses, nonlinear interactions can additionally be harnessed and, by controlling the phase of the electric field of ultrashort laser pulses and employing line-by-line pulse shaping, the combination of coherent control over quantum-mechanical processes and high-sensitivity broadband frequency comb spectroscopy techniques can be envisioned. Frequency comb Fourier transform spectroscopy might establish the basis of ground-breaking spectroscopic tools and open up new insights into the understanding of the structure of matter, as well

as new horizons in advanced diagnostic instruments in chemistry and biomedicine.

Future applications. Beyond enabling advances in molecular spectroscopy, mid-infrared frequency combs are expected not only to add tremendous value to existing applications, but also to have a strong impact on emerging or unexpected fields throughout many areas of research.

For instance, the line-by-line pulse shaping⁹⁷ of mid-infrared frequency combs would allow for the coherent control over molecular vibrational excitation in the electronic ground state or elementary excitations in condensed matter systems. Traditional shaping devices are either not transparent or exhibit very low diffraction efficiency in this wavelength region, which obstructs the direct modulation of the electric field transients. One approach may be to shape the signal beam of an OPO-based frequency comb.

Frequency combs in the near-infrared and visible regions have recently enabled great improvements^{98,99} in the calibration of ground-based astronomical spectrographs. The atmospheric ‘windows’ in the mid-infrared are beneficial to the radial velocity technique¹⁰⁰ for detecting high red-shift galaxies and exoplanets, as well as the spectroscopic characterization of their atmospheres. Several existing or planned mid-infrared cross-dispersed echelle spectrographs at large telescope facilities, such as the CRYogenic high-resolution InfraRed Echelle Spectrograph¹⁰¹ at the Very Large Telescope, may benefit from calibration using a mid-infrared frequency comb with a large mode spacing.

Mid-infrared frequency combs can serve as pump sources for extending the spectral territories covered by frequency comb techniques. Femtosecond thulium fibre lasers are already used⁵¹ for synchronously pumping OPOs that span the 3–6 μm range. Access to mid-infrared wavelengths also extends^{102,103} the cut-off energy in high-harmonic generation, which provides intriguing prospects for ultrafast phenomena, the generation of attosecond pulses and the extension of precision spectroscopy into the extreme ultraviolet region. Moreover, as sources of phase-stabilized femtosecond pulses, frequency combs have made an essential contribution to the production of attosecond pulses and the complete recovery of electric field transients. Intense few-cycle long-wavelength laser systems¹⁰⁴ with carrier-envelope phase stabilization may therefore benefit high-field physics applications.

Many other applications might also emerge in the near future, such as dual-comb static optical coherence tomography¹⁰⁵ of tissues with low water content, or the dual-comb calibration¹⁰⁶ of rapidly swept CW mid-infrared lasers.

Conclusion

Over the past decade, near-infrared and visible frequency combs have revolutionized optical frequency metrology and optical frequency synthesis. The rapidly evolving developments of frequency comb generator technology in the mid-infrared spectral region suggest that mature and turn-key sources will soon become available, which will stimulate much interest in novel applications. However, fundamental and technological progress in several key areas is still yet to be achieved. Direct laser emission of few-cycle phase coherent pulses across the mid-infrared range, improved materials for efficient nonlinear frequency conversion over a broad spectral bandwidth beyond 6 μm and octave-spanning low-phase noise microresonator-based frequency combs centred deeper in the mid-infrared are among the most important goals of the field. Current applications, mostly benefiting the spectroscopy of molecules, have opened up intriguing prospects in frequency metrology and trace-gas detection. As the performance frontiers of these photonic tools are advanced, we hope to see the development of unforeseen groundbreaking measurement techniques and unexpected scientific discoveries.

References

1. Udem, T., Holzwarth, R. & Hänsch, T. W. Optical frequency metrology. *Nature* **416**, 233–237 (2002).
2. Cundiff, S. T. & Ye, J. Colloquium: Femtosecond optical frequency combs. *Rev. Mod. Phys.* **75**, 325–342 (2003).
3. Ye, J. & Cundiff, S. T. (eds) *Femtosecond Optical Frequency Comb: Principle, Operation and Applications* (Springer, 2005).
4. Hall, J. L. Nobel Lecture: Defining and measuring optical frequencies. *Rev. Mod. Phys.* **78**, 1279–1295 (2006).
5. Hänsch, T. W. Nobel Lecture: Passion for precision. *Rev. Mod. Phys.* **78**, 1297–1309 (2006).
6. Diddams, S. A. The evolving optical frequency comb. *J. Opt. Soc. Am. B* **27**, B51–B62 (2010).
7. Page, R. H. *et al.* Cr²⁺-doped zinc chalcogenides as efficient, widely tunable mid-infrared lasers. *IEEE J. Quant. Electron.* **33**, 609–619 (1997).
8. Sorokina, I. T. Crystalline mid-infrared lasers. *Top. Appl. Phys.* **89**, 255–349 (2003).
9. Sorokin, E., Naumov, S. & Sorokina, I. T. Ultrabroadband infrared solid-state lasers. *IEEE J. Sel. Top. Quant. Electron.* **11**, 690–712 (2005).
10. Mirov, S. B. *et al.* Progress in mid-IR Cr²⁺ and Fe²⁺ doped II–VI materials and lasers invited. *Opt. Mater. Express* **1**, 898–910 (2011).
11. Sorokin, E., Sorokina, I. T., Mandon, J., Guelachvili, G. & Picqué, N. Sensitive multiplex spectroscopy in the molecular fingerprint 2.4 μm region with a Cr²⁺:ZnSe femtosecond laser. *Opt. Express* **15**, 16540–16545 (2007).
12. Cizmeciyan, M. N., Cankaya, H., Kurt, A. & Sennaroglu, A. Kerr-lens mode-locked femtosecond Cr²⁺:ZnSe laser at 2420 nm. *Opt. Lett.* **34**, 3056–3058 (2009).
13. Slobodtchikov, E. & Moulton, P. Progress in ultrafast Cr:ZnSe lasers in *Lasers, Sources, and Related Photonic Devices* paper AW5A.4 (OSA, 2012).
14. Sorokin, E., Tolstik, N. & Sorokina, I. Kerr-lens mode-locked Cr:ZnS laser in *Lasers, Sources, and Related Photonic Devices* paper AW5A.5 (OSA, 2012).
15. Bernhardt, B. *et al.* Mid-infrared dual-comb spectroscopy with 2.4 μm Cr²⁺:ZnSe femtosecond lasers. *Appl. Phys. B* **100**, 3–8 (2010).
16. Fedorov, V. V. *et al.* 3.77–5.05 μm tunable solid-state lasers based on Fe²⁺-doped ZnSe crystals operating at low and room temperatures. *IEEE J. Quant. Electron.* **42**, 907–917 (2006).
17. Frolov, M. P. *et al.* Laser radiation tunable within the range of 4.35–5.45 μm in a ZnTe crystal doped with Fe²⁺ ions. *J. Russ. Laser. Res.* **32**, 528–536 (2011).
18. Kozlovsky, V. I. *et al.* Pulsed Fe²⁺:ZnS laser continuously tunable in the wavelength range of 3.49–4.65 μm . *Quantum Electron.* **41**, 1–3 (2011).
19. Pollnau, M. & Jackson, S. D. in *Topics in Applied Physics* Vol. 89 (eds Sorokina, I. & Vodopyanov, K.) 219–255 (Springer, 2003).
20. Nelson, L. E., Ippen, E. P. & Haus, H. A. Broadly tunable sub-500 fs pulses from an additive-pulse mode-locked thulium-doped fiber ring laser. *Appl. Phys. Lett.* **67**, 19–21 (1995).
21. Solodyankin, M. A. *et al.* Mode-locked 1.93 μm thulium fiber laser with a carbon nanotube absorber. *Opt. Lett.* **33**, 1336–1338 (2008).
22. Kieu, K. & Wise, F. W. Soliton thulium-doped fiber laser with carbon nanotube saturable absorber. *IEEE Photon. Tech. Lett.* **21**, 128–130 (2009).
23. Haxsen, F., Wandt, D., Morgner, U., Neumann, J. & Kracht, D. Pulse characteristics of a passively mode-locked thulium fiber laser with positive and negative cavity dispersion. *Opt. Express* **18**, 18981–18988 (2010).
24. Wang, Q., Geng, J. H., Jiang, Z., Luo, T. & Jiang, S. B. Mode-locked Tm–Ho-codoped fiber laser at 2.06 μm . *IEEE Photon. Tech. Lett.* **23**, 682–684 (2011).
25. Phillips, C. R. *et al.* Supercontinuum generation in quasi-phase-matched LiNbO₃ waveguide pumped by a Tm-doped fiber laser system. *Opt. Lett.* **36**, 3912–3914 (2011).
26. Adler, F. & Diddams, S. A. High-power, hybrid Er:fiber/Tm:fiber frequency comb source in the 2 μm wavelength region. *Opt. Lett.* **37**, 1400–1402 (2012).
27. Coluccelli, N. *et al.* 1.6-W self-referenced frequency comb at 2.06 μm using a Ho:YLF multipass amplifier. *Opt. Lett.* **36**, 2299–2301 (2011).
28. Hofstetter, D. & Faist, J. High performance quantum cascade lasers and their applications. *Top. Appl. Phys.* **89**, 61–96 (2003).
29. Paiella, R. *et al.* Self-mode-locking of quantum cascade lasers with giant ultrafast optical nonlinearities. *Science* **290**, 1739–1742 (2000).
30. Wang, C. Y. *et al.* Mode-locked pulses from mid-infrared quantum cascade lasers. *Opt. Express* **17**, 12929–12943 (2009).
31. Fischer, C. & Sigrist, M. W. Mid-IR difference frequency generation. *Top. Appl. Phys.* **89**, 97–140 (2003).
32. Maddaloni, P., Malara, P., Gagliardi, G. & De Natale, P. Mid-infrared fibre-based optical comb. *New J. Phys.* **8**, 262 (2006).
33. Baumann, E. *et al.* Spectroscopy of the methane ν_3 band with an accurate midinfrared coherent dual-comb spectrometer. *Phys. Rev. A* **84**, 062513 (2011).
34. Foreman, S. M., Jones, D. J. & Ye, J. Flexible and rapidly configurable femtosecond pulse generation in the mid-IR. *Opt. Lett.* **28**, 370–372 (2003).
35. Foreman, S. M. *et al.* Demonstration of a HeNe/CH₄-based optical molecular clock. *Opt. Lett.* **30**, 570–572 (2005).
36. Erny, C. *et al.* Mid-infrared difference-frequency generation of ultrashort pulses tunable between 3.2 and 4.8 μm from a compact fiber source. *Opt. Lett.* **32**, 1138–1140 (2007).
37. Gubin, M. A. *et al.* Femtosecond fiber laser based methane optical clock. *Appl. Phys. B* **95**, 661–666 (2009).
38. Sell, A., Scheu, R., Leitenstorfer, A. & Huber, R. Field-resolved detection of phase-locked infrared transients from a compact Er:fiber system tunable between 55 and 107 THz. *Appl. Phys. Lett.* **93**, 251107 (2008).
39. Keilmann, F., Gohle, C. & Holzwarth, R. Time-domain mid-infrared frequency-comb spectrometer. *Opt. Lett.* **29**, 1542–1544 (2004).
40. Schliesser, A., Brehm, M., Keilmann, F. & van der Weide, D. W. Frequency-comb infrared spectrometer for rapid, remote chemical sensing. *Opt. Express* **13**, 9029–9038 (2005).
41. Gambetta, A., Ramponi, R. & Marangoni, M. Mid-infrared optical combs from a compact amplified Er-doped fiber oscillator. *Opt. Lett.* **33**, 2671–2673 (2008).
42. Keilmann, F. & Amarie, S. Mid-infrared frequency comb spanning an octave based on an Er fiber laser and difference-frequency generation. *J. Infrared Millim. Te.* **33**, 479–484 (2012).
43. Ruehl, A. *et al.* Widely-tunable mid-IR frequency comb source based on difference frequency generation. *Opt. Lett.* **37**, 2232–2234 (2012).
44. Ebrahimzadeh, M. in *Topics in Applied Physics* Vol. 89 (eds Sorokina, I. & Vodopyanov, K.) 179–218 (Springer, 2003).
45. Vodopyanov, K. in *Topics in Applied Physics* Vol. 89 (eds Sorokina, I. & Vodopyanov, K.) 141–178 (Springer, 2003).
46. Reid, D. T., Gale, B. J. S. & Sun, J. Frequency comb generation and carrier-envelope phase control in femtosecond optical parametric oscillators. *Laser Phys.* **18**, 87–103 (2008).
47. Sun, J. H., Gale, B. J. S. & Reid, D. T. Composite frequency comb spanning 0.4–2.4 μm from a phase-controlled femtosecond Ti:sapphire laser and synchronously pumped optical parametric oscillator. *Opt. Lett.* **32**, 1414–1416 (2007).
48. Adler, F. *et al.* Phase-stabilized, 1.5 W frequency comb at 2.8–4.8 μm . *Opt. Lett.* **34**, 1330–1332 (2009).
49. Wong, S. T., Vodopyanov, K. L. & Byer, R. L. Self-phase-locked divide-by-2 optical parametric oscillator as a broadband frequency comb source. *J. Opt. Soc. Am. B* **27**, 876–882 (2010).
50. Leindecker, N., Marandi, A., Byer, R. L. & Vodopyanov, K. L. Broadband degenerate OPO for mid-infrared frequency comb generation. *Opt. Express* **19**, 6304–6310 (2011).
51. Leindecker, N. *et al.* Octave-spanning ultrafast OPO with 2.6–6.1 μm instantaneous bandwidth pumped by femtosecond Tm-fiber laser. *Opt. Express* **20**, 7046–7053 (2012).
52. Vodopyanov, K. L., Sorokin, E., Sorokina, I. T. & Schunemann, P. G. Mid-IR frequency comb source spanning 4.4–5.4 μm based on subharmonic GaAs optical parametric oscillator. *Opt. Lett.* **36**, 2275–2277 (2011).
53. Savchenkov, A. A. *et al.* Low threshold optical oscillations in a whispering gallery mode CaF₂ resonator. *Phys. Rev. Lett.* **93**, 243905 (2004).
54. Del'Haye, P. *et al.* Optical frequency comb generation from a monolithic microresonator. *Nature* **450**, 1214–1217 (2007).
55. Kippenberg, T. J., Spillane, S. M. & Vahala, K. J. Kerr-nonlinearity optical parametric oscillation in an ultrahigh-Q toroid microcavity. *Phys. Rev. Lett.* **93**, 083904 (2004).
56. Del'Haye, P. *et al.* Octave spanning tunable frequency comb from a microresonator. *Phys. Rev. Lett.* **107**, 063901 (2011).
57. Okawachi, Y. *et al.* Octave-spanning frequency comb generation in a silicon nitride chip. *Opt. Lett.* **36**, 3398–3400 (2011).
58. Domachuk, P. *et al.* Over 4000 nm bandwidth of mid-IR supercontinuum generation in sub-centimeter segments of highly nonlinear tellurite PCFs. *Opt. Express* **16**, 7161–7168 (2008).
59. Yeom, D. I. *et al.* Low-threshold supercontinuum generation in highly nonlinear chalcogenide nanowires. *Opt. Lett.* **33**, 660–662 (2008).
60. Kuyken, B. *et al.* Mid-infrared to telecom-band supercontinuum generation in highly nonlinear silicon-on-insulator wire waveguides. *Opt. Express* **19**, 20172–20181 (2011).
61. Qin, G. S. *et al.* Wideband supercontinuum generation in tapered tellurite microstructured fibers. *Laser Phys.* **21**, 1115–1121 (2011).
62. Herr, T. *et al.* Universal dynamics of Kerr frequency comb formation in microresonators. Preprint at <http://arxiv.org/abs/1111.3071> (2011).
63. Savchenkov, A. A. *et al.* Tunable optical frequency comb with a crystalline whispering gallery mode resonator. *Phys. Rev. Lett.* **101**, 093902 (2008).

64. Wang, C. Y. *et al.* Mid-infrared frequency combs based on microresonators. Preprint at <http://arxiv.org/abs/1109.2716> (2011).
65. Daussy, C. *et al.* Long-distance frequency dissemination with a resolution of 10^{-17} . *Phys. Rev. Lett.* **94**, 203904 (2005).
66. Shelkovich, A., Butcher, R. J., Chardonnet, C. & Amy-Klein, A. Stability of the proton-to-electron mass ratio. *Phys. Rev. Lett.* **100**, 150801 (2008).
67. Lemarchand, C. *et al.* Progress towards an accurate determination of the Boltzmann constant by Doppler spectroscopy. *New J. Phys.* **13**, 073028 (2011).
68. Darquié, B. *et al.* Progress toward the first observation of parity violation in chiral molecules by high-resolution laser spectroscopy. *Chirality* **22**, 870–884 (2010).
69. Malara, P., Maddaloni, P., Gagliardi, G. & De Natale, P. Absolute frequency measurement of molecular transitions by a direct link to a comb generated around 3 μm . *Opt. Express* **16**, 8242–8249 (2008).
70. Amy-Klein, A. *et al.* Absolute frequency measurement of a SF_6 two-photon line by use of a femtosecond optical comb and sum-frequency generation. *Opt. Lett.* **30**, 3320–3322 (2005).
71. Bielsa, F. *et al.* HCOOH high-resolution spectroscopy in the 9.18 μm region. *J. Mol. Spectrosc.* **247**, 41–46 (2008).
72. Gatti, D. *et al.* High-precision molecular interrogation by direct referencing of a quantum-cascade-laser to a near-infrared frequency comb. *Opt. Express* **19**, 17520–17527 (2011).
73. Bartalini, S. *et al.* Frequency-comb-referenced quantum-cascade laser at 4.4 μm . *Opt. Lett.* **32**, 988–990 (2007).
74. Giusfredi, G. *et al.* Saturated-absorption cavity ring-down spectroscopy. *Phys. Rev. Lett.* **104**, 110801 (2010).
75. Okubo, S., Nakayama, H., Iwakuni, K., Inaba, H. & Sasada, H. Absolute frequency list of the ν_3 -band transitions of methane at a relative uncertainty level of 10^{-11} . *Opt. Express* **19**, 23878–23888 (2011).
76. Vainio, M., Merimaa, M. & Halonen, L. Frequency-comb-referenced molecular spectroscopy in the mid-infrared region. *Opt. Lett.* **36**, 4122–4124 (2011).
77. Marian, A., Stowe, M. C., Lawall, J. R., Felinto, D. & Ye, J. United time-frequency spectroscopy for dynamics and global structure. *Science* **306**, 2063–2068 (2004).
78. Teets, R., Eckstein, J. & Hänsch, T. W. Coherent two-photon excitation by multiple light pulses. *Phys. Rev. Lett.* **38**, 760–764 (1977).
79. Eckstein, J. N., Ferguson, A. I. & Hänsch, T. W. High-resolution two-photon spectroscopy with picosecond light pulses. *Phys. Rev. Lett.* **40**, 847–850 (1978).
80. Diddams, S. A., Hollberg, L. & Mbele, V. Molecular fingerprinting with the resolved modes of a femtosecond laser frequency comb. *Nature* **445**, 627–630 (2007).
81. Mandon, J., Guelachvili, G. & Picqué, N. Fourier transform spectroscopy with a laser frequency comb. *Nature Photon.* **4**, 55–57 (2009).
82. Bernhardt, B. *et al.* Cavity-enhanced dual-comb spectroscopy. *Nature Photon.* **4**, 55–57 (2010).
83. Thorpe, M. J. & Ye, J. Cavity-enhanced direct frequency comb spectroscopy. *Appl. Phys. B* **91**, 397–414 (2008).
84. Adler, F. *et al.* Cavity-enhanced direct frequency comb spectroscopy: technology and applications. *Ann. Rev. Anal. Chem.* **3**, 175–205 (2010).
85. Thorpe, M. J. *et al.* Broadband cavity ringdown spectroscopy for sensitive and rapid molecular detection. *Science* **311**, 1595–1599 (2006).
86. Griffiths, P. R. & De Haseth, J. A. *Fourier Transform Infrared Spectroscopy* 2nd edn, 1–656 (Wiley, 2007).
87. Adler, F. *et al.* Mid-infrared Fourier transform spectroscopy with a broadband frequency comb. *Opt. Express* **18**, 21861–21872 (2010).
88. Foltynowicz, A., Malowski, P., Fleisher, A. J., Bjork, B. & Ye, J. Cavity-enhanced optical frequency comb spectroscopy in the mid-infrared — application to trace detection of H_2O_2 . Preprint at <http://arxiv.org/abs/1202.1216> (2012).
89. Amarie, S. & Keilmann, F. Broadband-infrared assessment of phonon resonance in scattering-type near-field microscopy. *Phys. Rev. B* **83**, 045404 (2011).
90. Ganz, T., Brehm, M., von Ribbeck, H. G., van der Weide, D. W. & Keilmann, F. Vector frequency-comb Fourier-transform spectroscopy for characterizing metamaterials. *New J. Phys.* **10**, 123007 (2008).
91. Brehm, M., Schliesser, A. & Keilmann, F. Spectroscopic near-field microscopy using frequency combs in the mid-infrared. *Opt. Express* **14**, 11222–11233 (2006).
92. Coddington, I., Swann, W. C. & Newbury, N. R. Coherent multiheterodyne spectroscopy using stabilized optical frequency combs. *Phys. Rev. Lett.* **100**, 013902 (2007).
93. Zolot, A. M. *et al.* Direct-comb molecular spectroscopy with accurate, resolved comb teeth over 43 THz. *Opt. Lett.* **37**, 638–640 (2012).
94. Ideguchi, T., Poisson, A., Guelachvili, G., Picqué, N. & Hänsch, T. W. Adaptive real-time dual-comb spectroscopy. Preprint at <http://arxiv.org/abs/1201.4177> (2012).
95. Ideguchi, T., Bernhardt, B., Guelachvili, G., Hänsch, T. W. & Picqué, N. Femtosecond stimulated raman dual-comb spectroscopy in CLEO: Applications and Technology paper CThC5.6 (OSA, 2012).
96. Zhang, Z. *et al.* Asynchronous midinfrared ultrafast optical parametric oscillator for dual-comb spectroscopy. *Opt. Lett.* **37**, 187–189 (2012).
97. Cundiff, S. T. & Weiner, A. M. Optical arbitrary waveform generation. *Nature Photon.* **4**, 760–766 (2010).
98. Steinmetz, T. *et al.* Laser frequency combs for astronomical observations. *Science* **23**, 1335–1337 (2008).
99. Li, C.-H. *et al.* A laser frequency comb that enables radial velocity measurements with a precision of 1 cm s^{-1} . *Nature* **452**, 610–612 (2008).
100. Mayor, M. & Queloz, D. A Jupiter-mass companion to a solar-type star. *Nature* **378**, 355–359 (1995).
101. Figueira, P. *et al.* Radial velocities with CRILES. Pushing precision down to 5–10 m/s. *Astron. Astrophys.* **511**, A55 (2010).
102. Sheehy, B. *et al.* High harmonic generation at long wavelengths. *Phys. Rev. Lett.* **83**, 5270–5273 (1999).
103. Krause, J. L., Schafer, K. J. & Kulander, K. C. High-order harmonic generation from atoms and ions in the high intensity regime. *Phys. Rev. Lett.* **68**, 3535–3538 (1992).
104. Silva, F., Bates, P. K., Esteban-Martin, A., Ebrahim-Zadeh, M. & Biegert, J. High-average-power, carrier-envelope phase-stable, few-cycle pulses at 2.1 μm from a collinear BiB_3O_6 optical parametric amplifier. *Opt. Lett.* **37**, 933–935 (2012).
105. Lee, S. J., Widiyatmoko, B., Kourogi, M. & Ohtsu, M. Ultrahigh scanning speed optical coherence tomography using optical frequency comb generators. *Jpn J. Appl. Phys.* **40**, L878–L880 (2001).
106. Giorgetta, F. R., Coddington, I., Baumann, E., Swann, W. C. & Newbury, N. R. Fast high-resolution spectroscopy of dynamic continuous-wave laser sources. *Nature Photon.* **4**, 853–857 (2010).
107. Rothman, L. S. *et al.* The HITRAN 2008 molecular spectroscopic database. *J. Quant. Spectrosc. Ra.* **110**, 533–572 (2009).

Acknowledgements

T.W.H. and N.P. acknowledge support by the European Associated Laboratory 'European Laboratory for Frequency Comb Spectroscopy' and the Max Planck Foundation. A.S. acknowledges support from a Marie Curie IAPP programme and the Swiss National Science Foundation. A. Amy-Klein, E. Baumann, B. Darquié, P. de Natale, A. Foltynowicz-Matyba, F. Keilmann, T.J. Kippenberg, D. Mazzotti, N.R. Newbury, K. Vodopyanov, C.Y. Wang and J. Ye are gratefully acknowledged for providing comments, data and figures.

Additional information

Correspondence and requests for materials should be addressed to N.P.